

MEC 203 MARS Integrity Rover Report

Group Number#47

Angelo Brescia; ID #117124776

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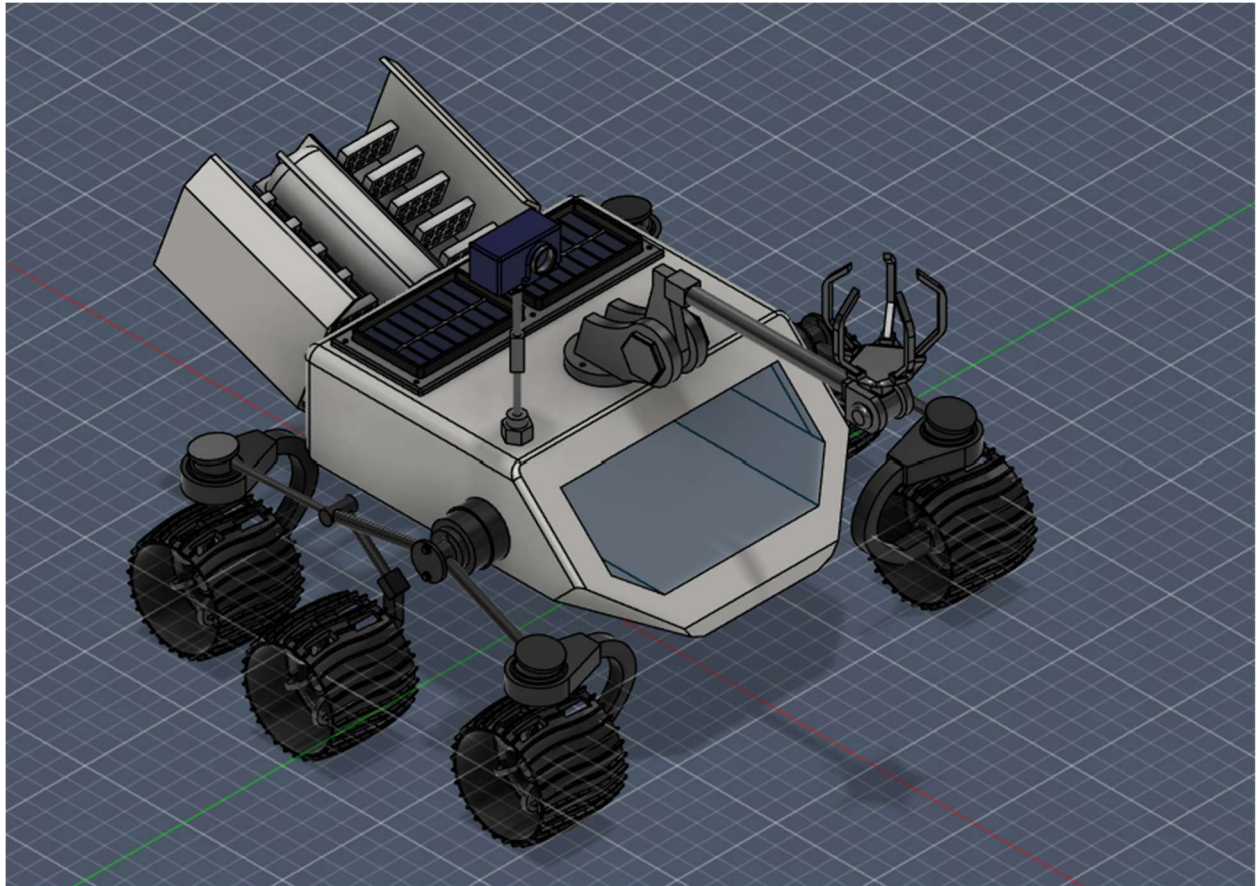


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MEC 203 Integrity Mars Rover

1 FOREWORD

I'm a designer for the NASA JPL Program and have been instructed on designing a new Mars Rover. This project involves the design and development of a sister rover for the Mars 2020 Perseverance mission. By utilizing the proven suspension and wheel design, the mission is to maximize reliability while testing the new gripper configurations for sample analysis.

2 PRODUCT DESIGN

2.1 Concept Design/Generation and Selection

The Preliminary Design phase began with a research period focused on industrial robotic arm configurations, specifically analyzing the trade-offs between operational stability and mechanical complexity. The concepts were visualized by observing how different grippers perform specific tasks and by examining the importance of joint placements in each design.



Concept Generation

Initially, the primary design concept featured a 2-point “grabber system” inspired by simple parallel-jaw grippers commonly found in robotics. The goal of this design involved minimizing the weight and overall movement of the system.

Selection and Justification

After evaluating the different ideas and prototypes, the two-grabber design was rejected as the analysis revealed 2 major flaws:

- 1) Grip strength: A two-point contact system is not sufficient for maintaining a solid and secure hold on irregularly shaped objects, leading to a high risk of slippage.
- 2) Stability: When researching the arms' physics, the two-grabber configuration is likely to create uneven torque distribution. Therefore, when under load, the grip will be imbalanced and can lead to mechanical failure

The Improved & Final Design:

To maximize grip reliability and versatility, the 5-point grabber system was produced. This design increased the contact surface area significantly and improved the arms' stability in high torque maneuvers. This design can handle all sorts of objects, as the arm can use the ends to secure an object or completely encapsulate the object using the 5 grippers.

2.2 Product specifications

The "Sister Rover" is designed for stability and high-performance sample collection in rough terrain. Below are the key specifications:

Chassis: High-strength frame with attached solar panels and camera mast.

Suspension: A Rocker-Bogie system that allows the wheels to climb over obstacles larger than the diameter of the wheel; Can traverse through rough terrain.

Arm System: A 5-point gripper designed to grab rocks of various sizes securely.

Not including the arm, the approximate dimension of the rover is 2.7x2.8x1.5 m

2.3 Details of the Design.

2.3.1 Why the Final Design Concept Was Selected?

Advantages:

1) Features

- 5-point gripper:

Unlike the Perseverance Rover, the 5-point gripper arm allows for the collection of various materials across the surface of Mars. This system allows an even torque distribution across the claw, ensuring much security when grabbing an irregular object.

- The Robotic Arm:

The arm itself had 2 pivot points. Both points lead to absolute precision when locating any item and can distribute the items at multiple points. The arm allows the claw not only to attain the ability to extract, but also to assemble

- The Chassis:

The shape of the chassis sits inside of the suspension and has been cut on the bottom to increase the clearance of the suspension. This protects the chassis from colliding with the suspension when crossing tough terrain. It is also cut at an angle in the front to allow the claw arm to rest properly, avoiding uneven weight distribution.

- MMRTG + Shields

The battery can supply the rover for many long periods of time. It will stay preserved as it is angled up, moving it farther from the ground. Its sides are also protected with shields, preventing dust and rubble from harming the sides of the battery

- Camera

The Camera can capture information in many different orientations. The camera can rotate a full 360 degrees, giving a complete view of the surroundings. The camera can also tilt down 90 degrees, which is essential for observing the body of the rover to check for any issues. It is also essential to inspect the ground directly in front of the rover before the arm moves to collect a sample. Each of these movements can be followed by moving the camera base down to 150mm to get the best capture for any given moment.

2) Cost Effectiveness

- The Chassis

The Chassis is not only cut to fit the suspension and help promote weight distribution but is also hollow. Material was saved while enhancing performance. It still holds stability as the weight is distributed properly and present in crucial areas, giving the rover an edge over the rest.

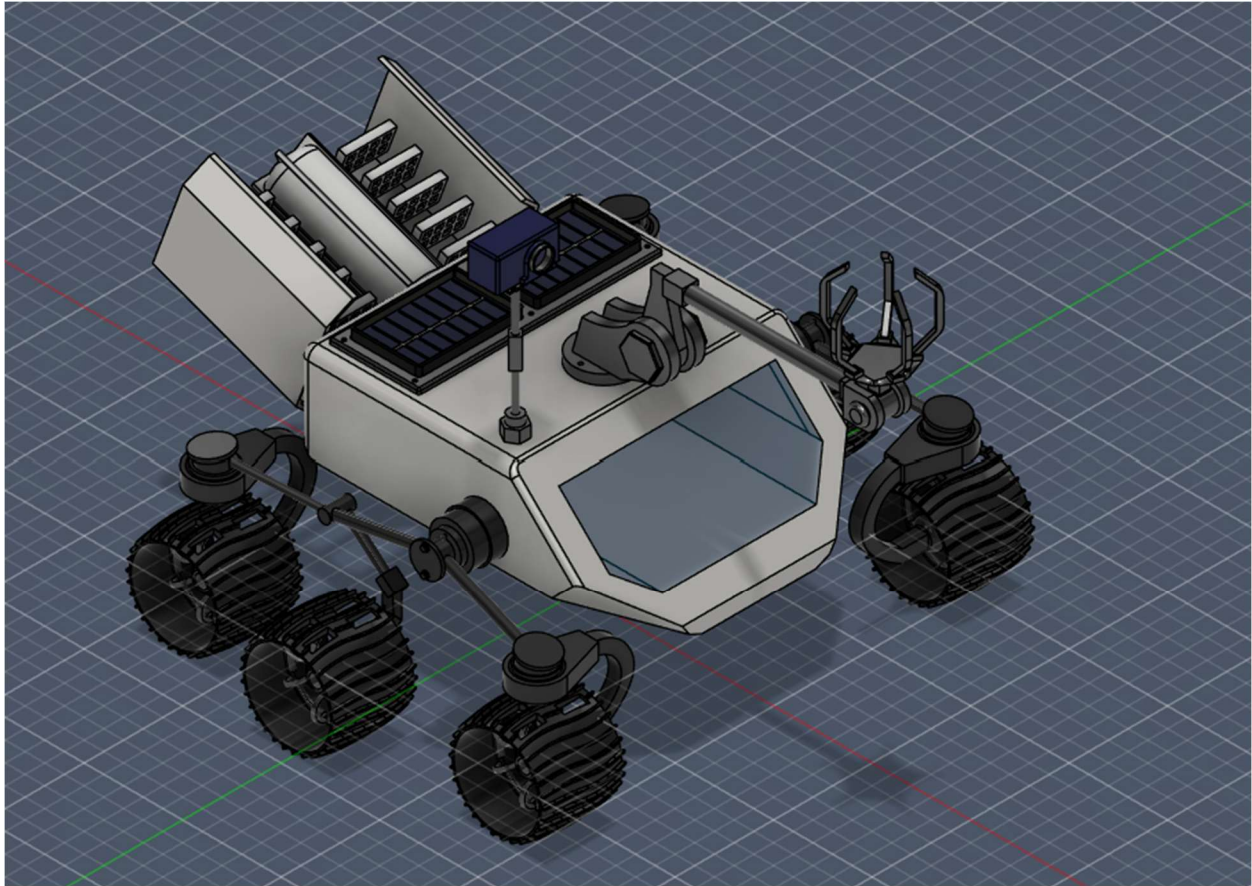
- Wheels

The outer edge of the wheels has a section where material has been cut out to prevent unnecessary use of material. The holes aren't very large to ensure no large material gets stuck inside, which can cause many problems to the rover on expeditions.

2.3.2 CAD Model

Picture and explanation of the entire assembled model here. All other components should be shown in the appendix.

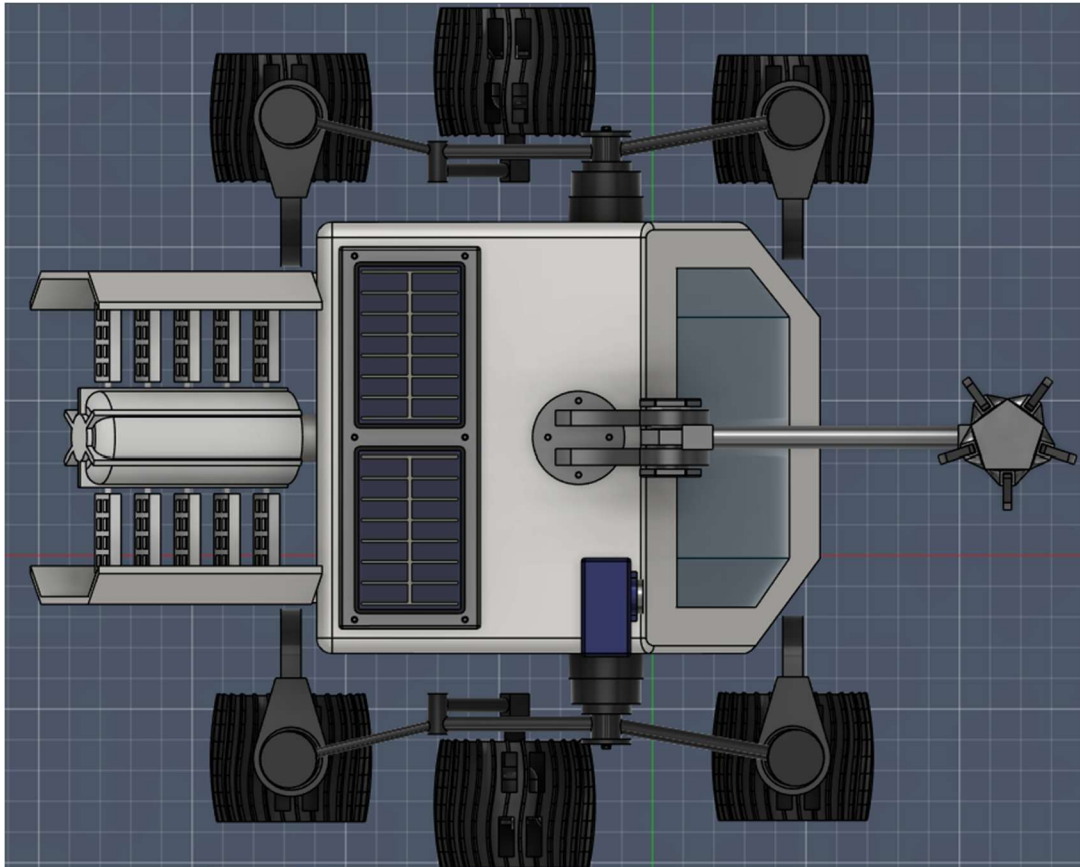
Pictorial View



Above is the rover in an isometric view. The chassis is the foundation of this rover in structural terms, as it is the hub for connecting components. The suspension sits at the sides off-center horizontally from the chassis so the wheels can be properly distributed to the true length of the entire rover (from this angle the suspension looks close to the bottom of the chassis but this can be seen better in the Rear-Side View) The outer wheels connect to the suspension from the inside, which contributes to a more flexible suspension that promotes traveling in rough terrain. The chassis allows the suspension to take on this shape as it is cut out in these areas to prevent overlapping or collision. The window in the front indicates the chassis is hollow, but it can still

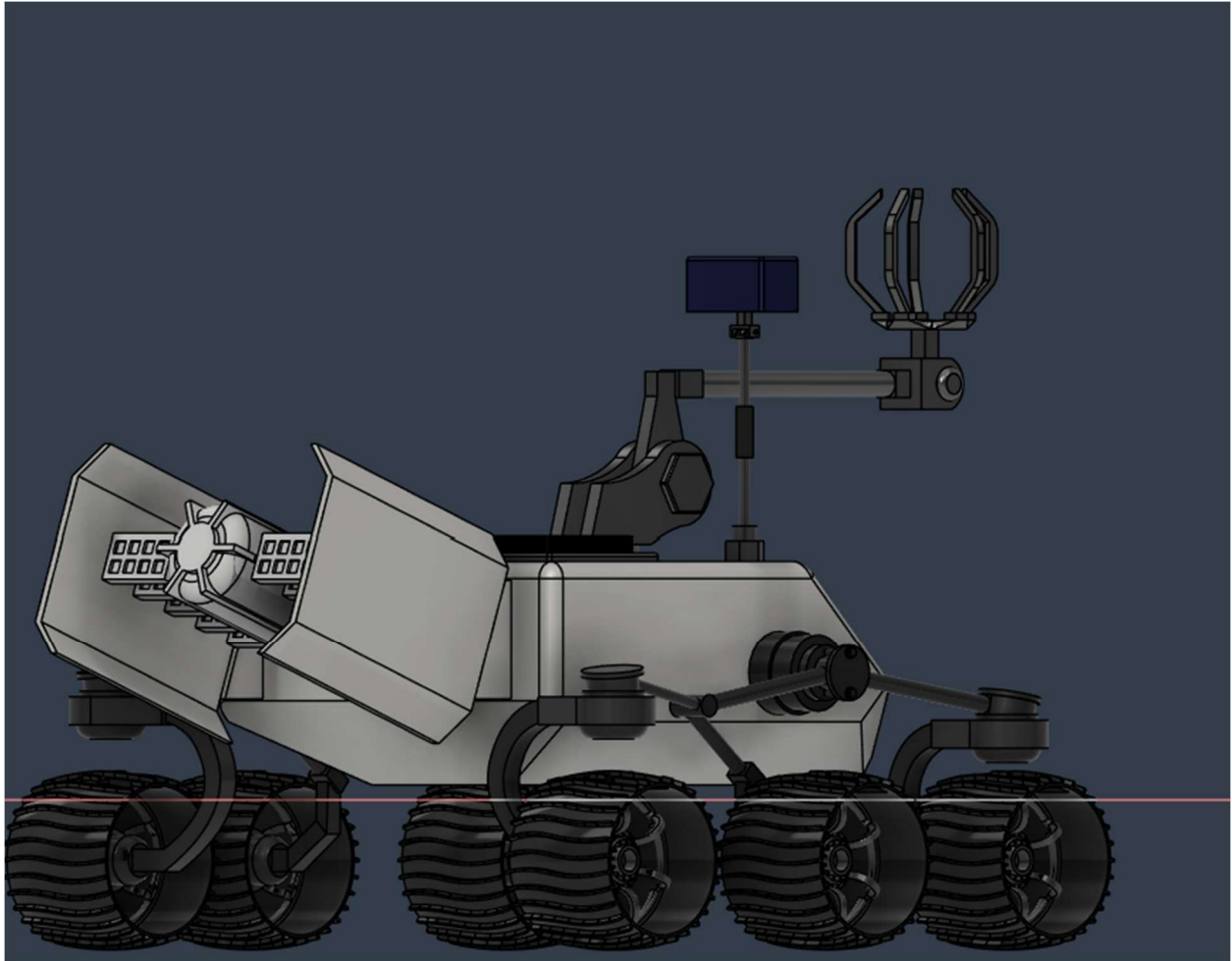
be seen as a strong base overall.

TOP VIEW



As seen in the top view, it is clear to see how the camera, solar panels, and arm sit on the chassis. The arm and panels both have cutouts where the head of the bolt will sit, and the bolt meets the thread inside the chassis. When observing the wheels, the cutouts can be seen clearly, which helps reduce unnecessary material and helps reduce the cost.

REAR SIDE VIEW



In the rear side view, the vertical alignment is seen more clearly as the true bottom of the rover is much farther below, as the chassis has been cut. The MMRTG sits in the back of the rover and is angled at 45 degrees, accompanied by 2 shield protectors. The battery will stay away from the ground as the arm and chassis help balance the weight. The two pivots of the arm can be seen from this view in their resting positions.

2.3.3 Tolerance and Fit

In this design, tolerances were applied to ensure the rover will move smoothly and with the motive for a simpler assembly. The specific “fits” were chosen and determined based on whether the components are supposed to move (pivots) or stay fixed (bolts).

Component	Bolt/Pin Size (mm)	Hole Size (mm)	Gap (mm)	Type of Fit
Arm Base Pivot	125	125.5	0.5	Precision Pivot
Claw Pivot	75	75.3	0.3	Precision Pivot
Suspension Bolt	15	16.5	1.5	Loose Clearance
Arm-to-Chassis	10	11	1	Standard Clearance
Solar Panel Bolt	8	9	1	Loose Clearance
Camera Pivot	7.5	8	0.5	Smooth Pivot

1) Moving Joints (The “Pivot” Fits)

- The Arm Base Pivot (125 mm): Hole Size: 125.5 mm

The 0.5mm gap will act as a precision fit; because the arm is considerably large, a precise fit is used to prevent significant wobbling during operation.

- Camera Pivot (7.5 mm): Hole Size: 8 mm

The gap ensures zero friction, which leads to easy tilting. This is essential because the camera should be used not only to observe the planet and its surroundings, but also to observe each component of the rover.

- Claw-to-arm Pivot (75mm): Hole Size: 75.3mm

Unlike the previous moving pivots with a 0.5mm clearance, a 0.3 clearance is used, giving it a more precise fit than the Arm Base Pivot. Since the claw is the main selling point of the rover, the tolerance needs to be more precise to minimize wobbling to ensure the 5-point gripper can align perfectly with small or more fragile samples.

2)

The Static Connections (Loose Fits)

- Solar Panel Bolts (8.00 mm): Hole Size: 9 mm

Reasoning: The solar panels require 6 mounting points. If the holes are too tight, and one hole is off by even a few millimeters, the second bolt will not fit in. A 1.0 mm gap is utilized to give wiggle room, allowing all bolts to align to minimize difficulty. The weight also justifies this fit as the panels are light compared to the other components attached to the rover.

Arm-to-Chassis Bolts (10.0 mm): Hole size: 11 mm

Reasoning: A medium clearance fit is desired so that the fit is tight enough to prevent shifting in the arm when tightened, but not too restrictive to the point where the bolts are forced into the thick chassis walls.

Suspension Bolts (15.0 mm): Hole Size: 16.5 mm

The suspension bolts can have a loose fit as each one is accompanied by a large 30mm washer. This washer allows there to be plenty of surface area for the long bolts to align with the corresponding holes. Although the Hole size may seem large, it is still small enough that any outside rubble or dust from Mars will not seep in and cause the bolt to seize.

3 REFERENCES

NASA Perseverance Rover 3d Model:

<https://science.nasa.gov/resource/mars-perseverance-rover-3d-model/>

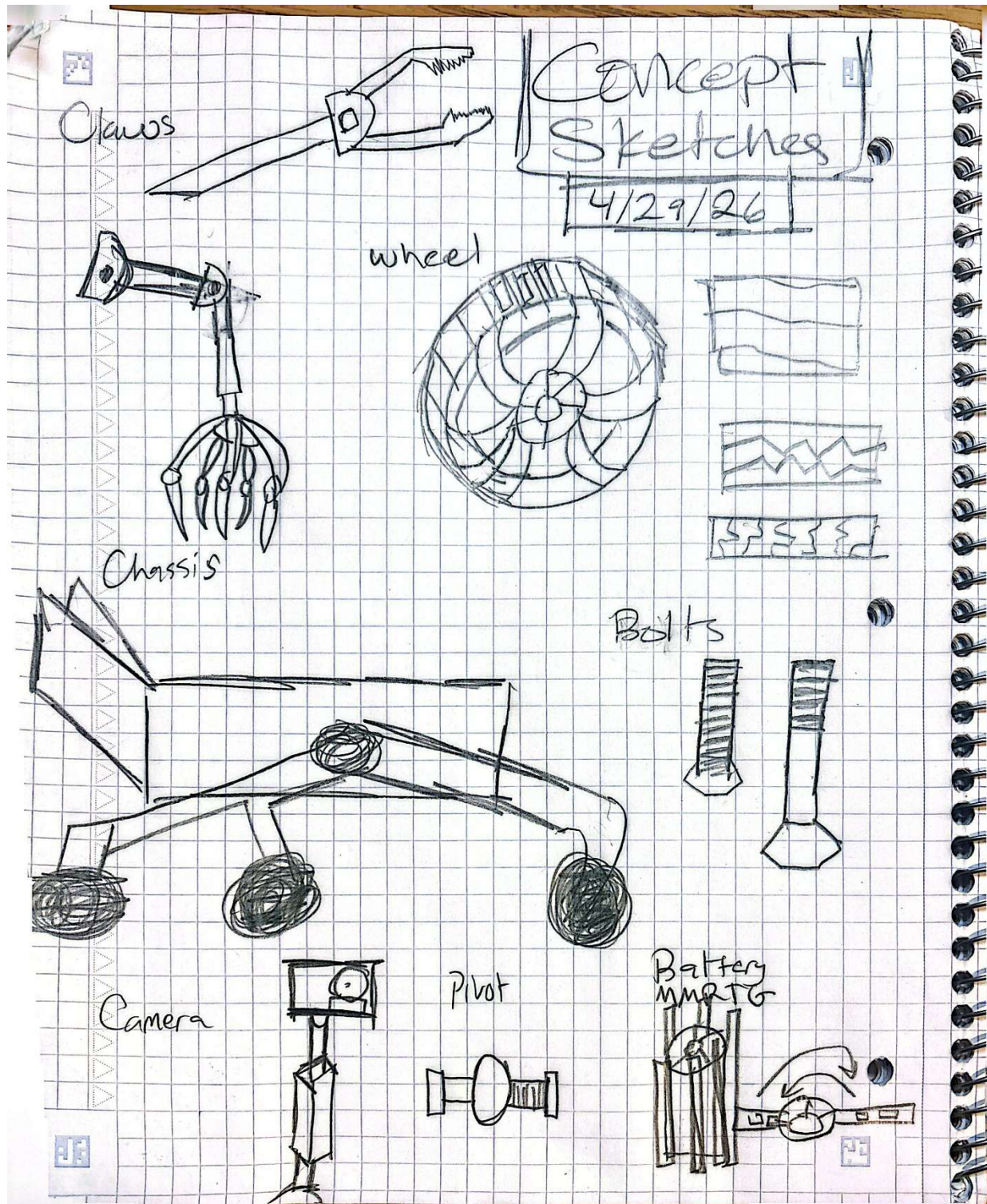
Claw Video

<https://www.youtube.com/watch?v=j4vn2uHj3Wg>

4 ACKNOWLEDGEMENTS

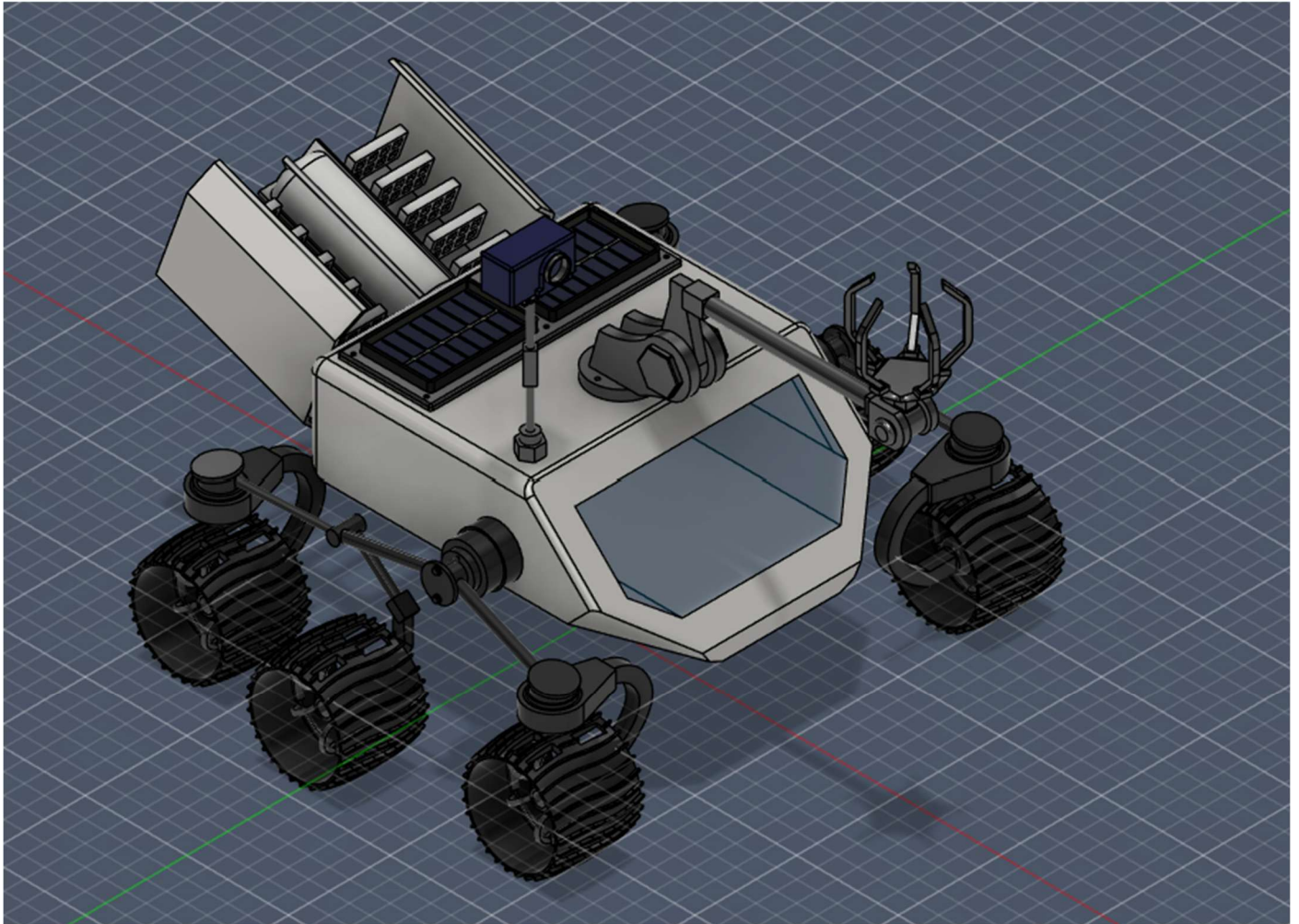
- Shehran Hasan: Student
- Satsang Sardar: Student

APPENDIX A CONCEPT HAND SKETCHES



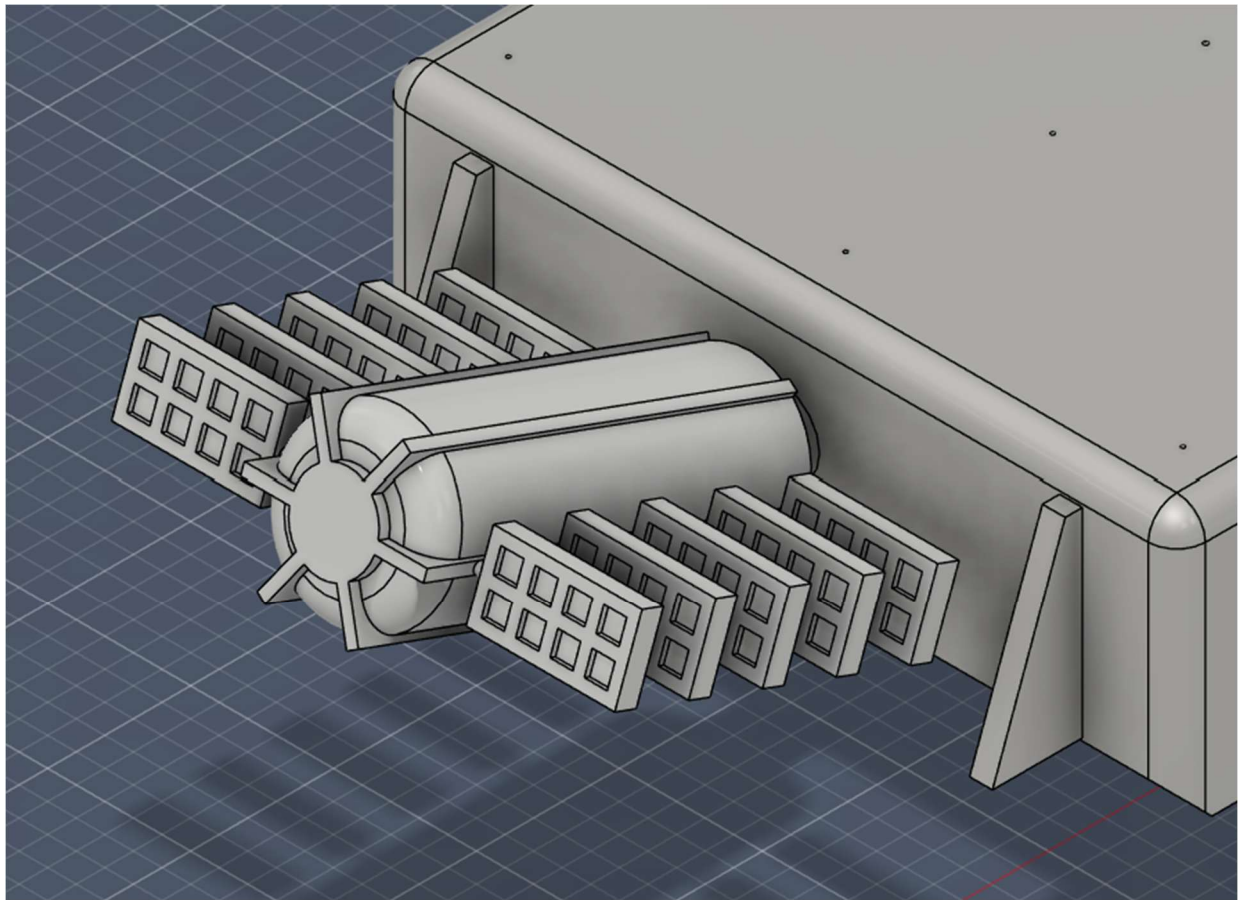
APPENDIX B 3D CAD MODELS AND DIMENSIONED DRAWINGS

B.1 Full assembled CAD Model

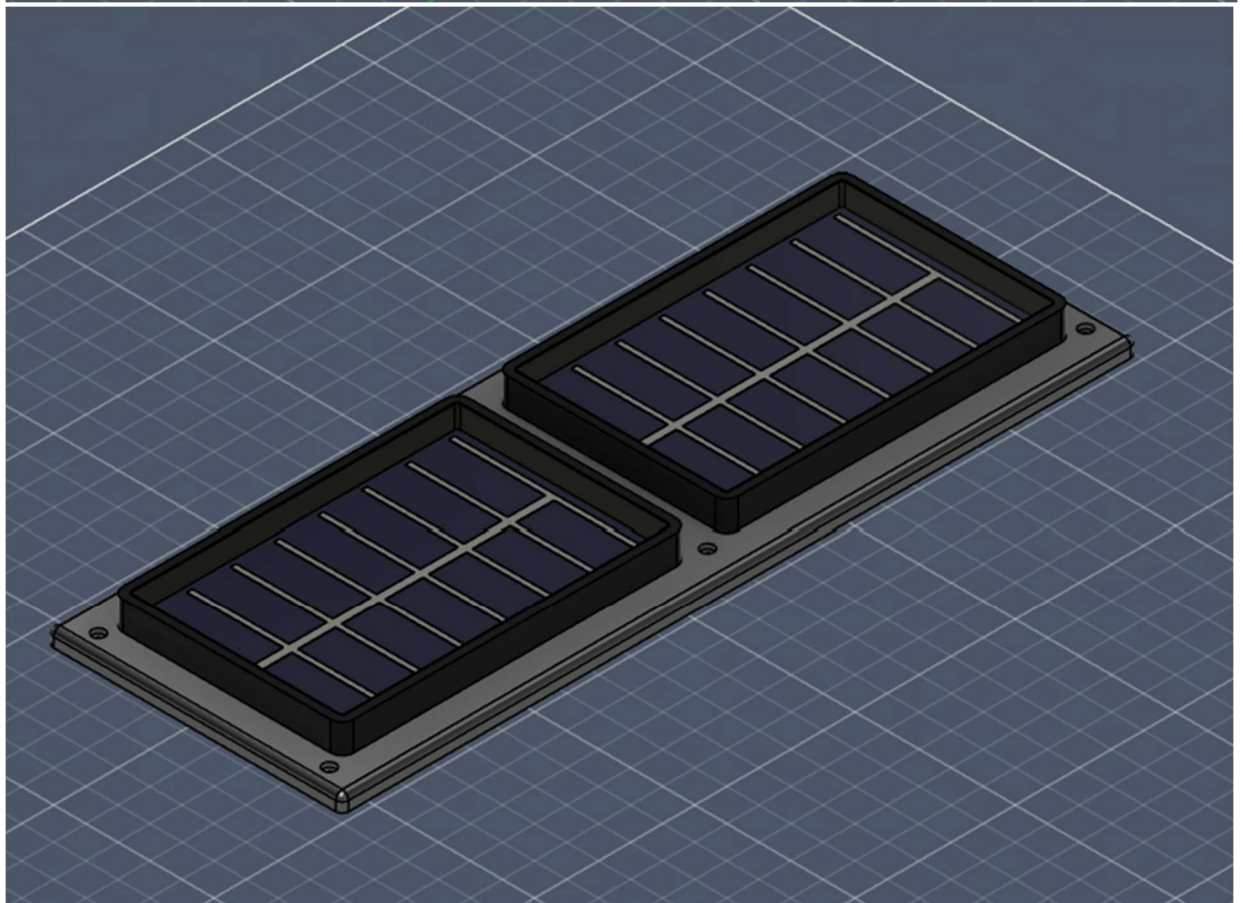
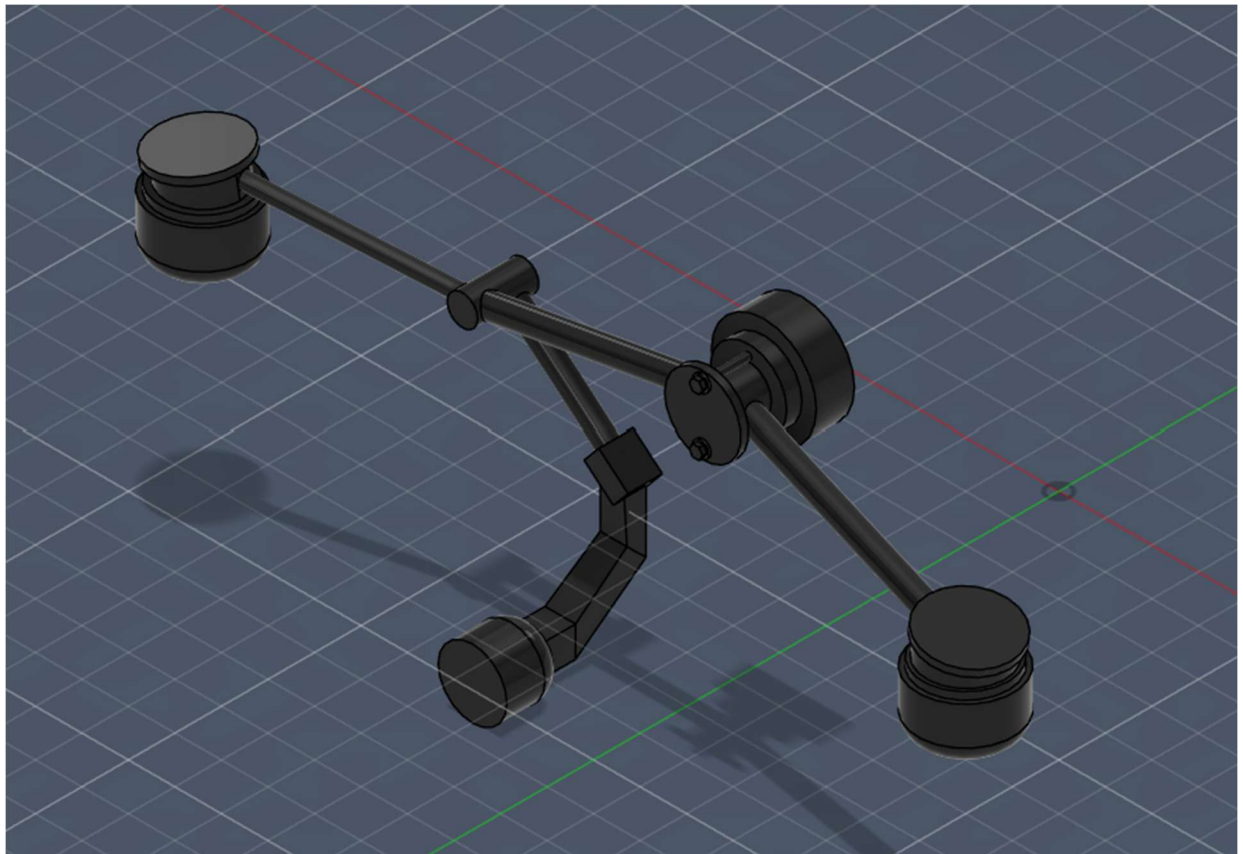


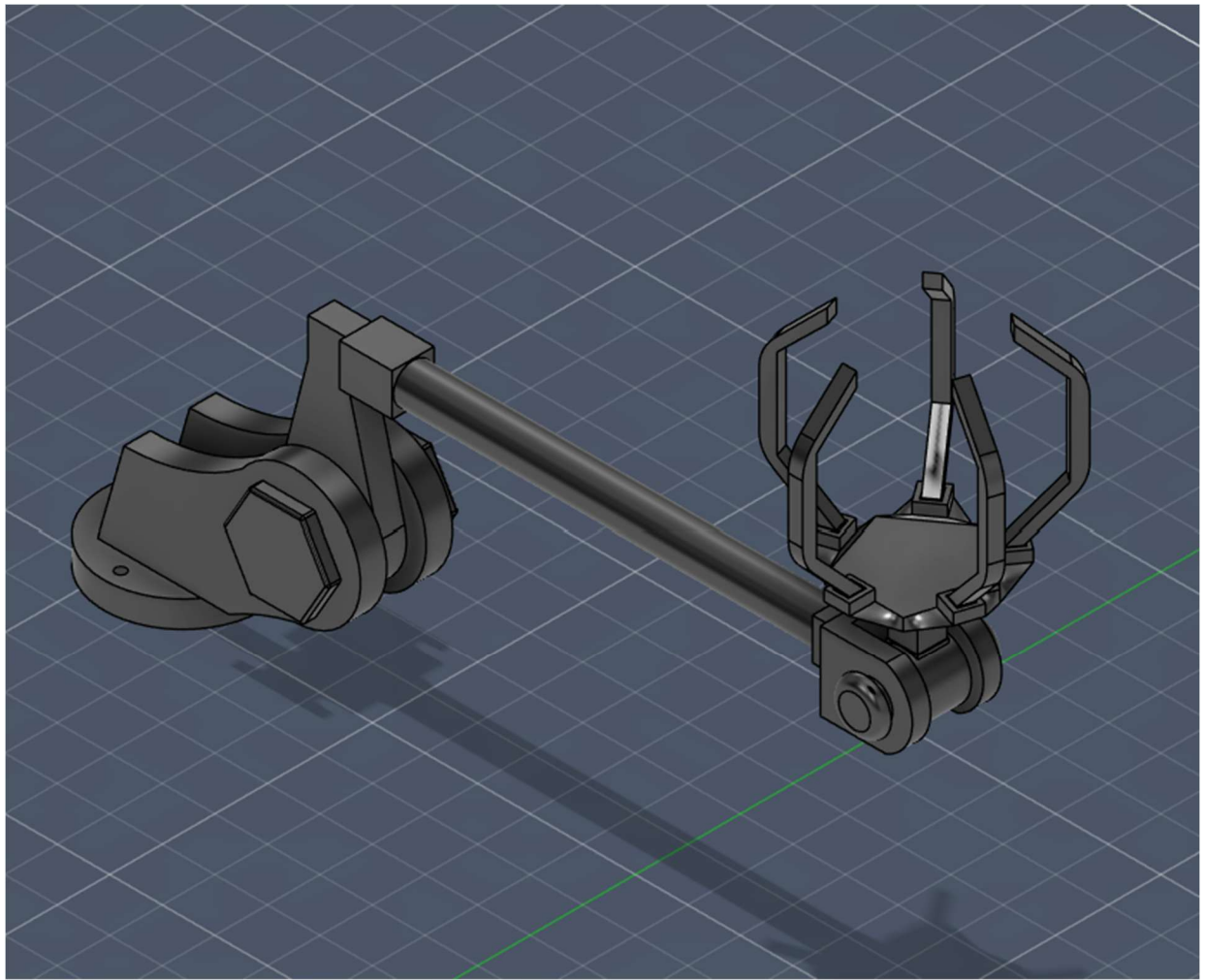
B.2 CAD Models of Individual Parts

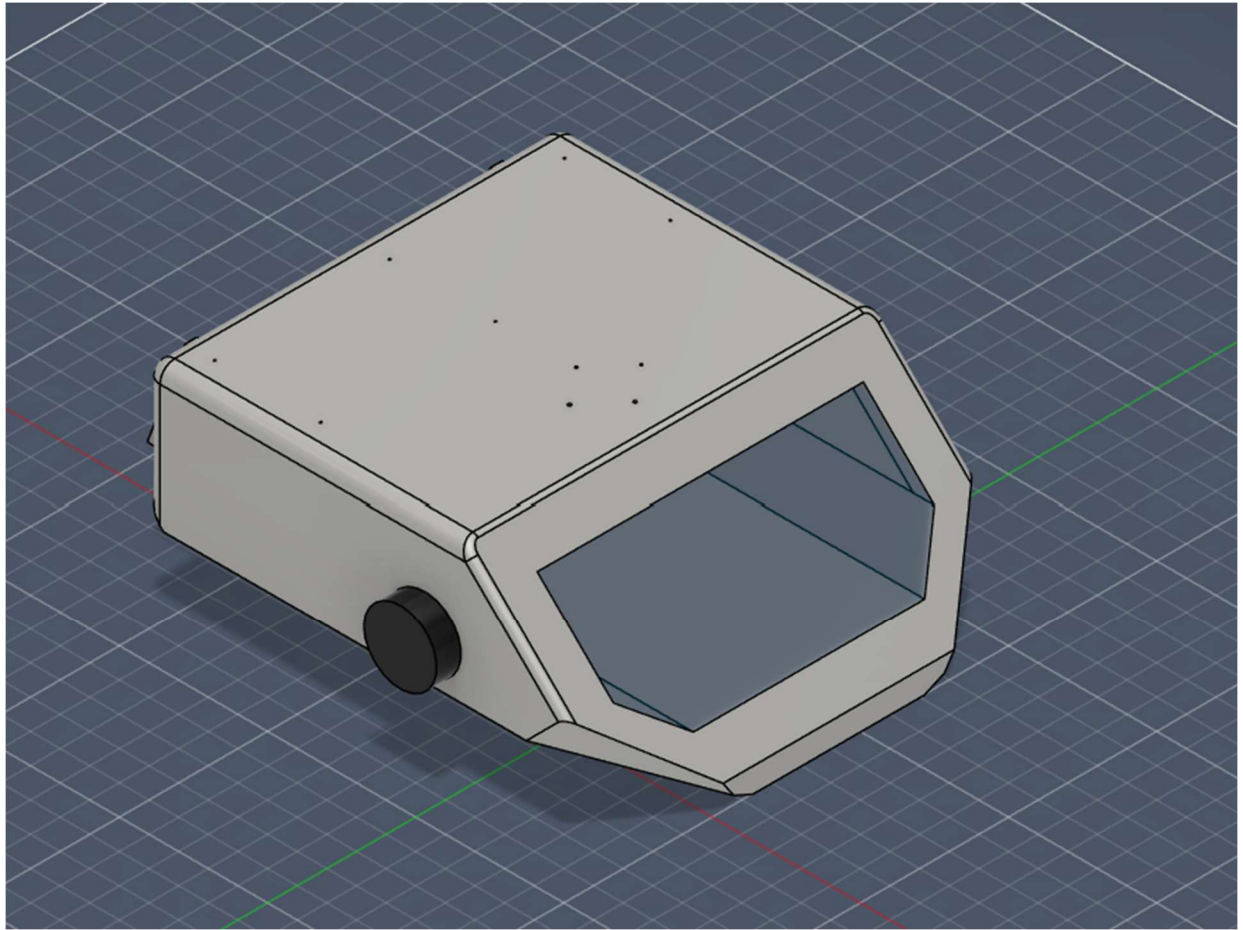












B.3 Dimensioned Drawings of Individual Parts

